

Profit, Loss & Discount — Solutions & Answers

Question 13

A man buys 35 kg of sugar and sets a marked price in order to make a 20% profit. He sells 5 kg at this price, and 15 kg at a 10% discount. Accidentally, 3 kg of sugar is wasted. He sells the remaining sugar by raising the marked price by p percent so as to make an overall profit of 15%. Then p is nearest to

- A) 22
- B) 35
- ✓ C) 25 ← Correct Answer
- D) 31

Solution:

Step 1: Calculate Cost Price (CP) and Marked Price (MP)

Let the CP of 1 kg of sugar be Rs. 100. Total CP of 35 kg = $35 \times 100 = \text{Rs. } 3500$.

The man wants to make a 20% profit initially, so the target SP = $3500 \times 1.20 = \text{Rs. } 4200$.

This is set as the Marked Price for 35 kg. MP per kg = $4200 / 35 = \text{Rs. } 120$.

Step 2: Calculate the revenue from initial sales

Sale 1: 5 kg at MP. SP1 = $5 \times 120 = \text{Rs. } 600$.

Sale 2: 15 kg at 10% discount. Discounted price = $120 \times 0.90 = \text{Rs. } 108$. SP2 = $15 \times 108 = \text{Rs. } 1620$.

Total revenue so far = $600 + 1620 = \text{Rs. } 2220$.

Step 3: Calculate the overall target SP and remaining quantity

Overall desired profit is 15%. Overall Target SP = $3500 \times 1.15 = \text{Rs. } 4025$.

Quantity sold = $5 + 15 = 20$ kg. Quantity wasted = 3 kg.

Remaining quantity to be sold = $35 - 20 - 3 = 12$ kg.

Step 4: Calculate the required SP for the remaining quantity

Required revenue from the remaining 12 kg = Overall Target SP - Revenue so far = $4025 - 2220 = \text{Rs. } 1805$.

Required SP per kg for the remaining sugar = $1805 / 12 = 150.42$.

Step 5: Calculate the percentage increase 'p'

New SP per kg = 150.42. Original MP per kg = 120. Increase in price = $150.42 - 120 = 30.42$.

Percentage increase $p = (\text{Increase} / \text{Original MP}) \times 100 = (30.42 / 120) \times 100 = 25.35\%$.

The nearest integer value for p is 25.

Correct Answer: C

Question 14

The selling price of a product is fixed to ensure 40% profit. If the product had cost 40% less and had been sold for 5 rupees less, then the resulting profit would have been 50%. The original

selling price, in rupees, of the product is

- 1) 15
- ✓ 2) 14 ← Correct Answer
- 3) 10
- 4) 20

Solution:

1. Define variables:

Let the original Cost Price (CP) be c . Let the original Selling Price (SP) be s .

2. Formulate the initial condition:

The SP is fixed to ensure a 40% profit. $s = c + 0.40c = 1.4c$ (Equation 1)

3. Formulate the new scenario:

The product had cost 40% less: New CP = $c - 0.40c = 0.6c$.

It had been sold for 5 rupees less: New SP = $s - 5$.

The resulting profit would have been 50%. New SP = New CP + 0.50(New CP) = $1.5 \times 0.6c = 0.9c$.

So: $s - 5 = 0.9c$ (Equation 2)

4. Solve the system of equations:

Substitute $s = 1.4c$ into Equation 2: $1.4c - 5 = 0.9c \rightarrow 0.5c = 5 \rightarrow c = 10$.

5. Find the original selling price:

$s = 1.4c = 1.4 \times 10 = 14$. The original selling price is Rs 14.

Correct Answer: 2

Question 15

Anil buys 12 toys and labels each with the same selling price. He sells 8 toys initially at 20% discount on the labeled price. Then he sells the remaining 4 toys at an additional 25% discount on the discounted price. Thus, he gets a total of Rs 2112, and makes a 10% profit. With no discounts, his percentage of profit would have been

- ✓ A) 50 ← Correct Answer
- B) 60
- C) 54
- D) 55

Solution:

1. Calculate the Cost Price (CP) per toy.

Total selling price (SP) = Rs 2112. Total profit = 10%.

Total CP = Total SP / (1 + Profit%) = $2112 / 1.10 = \text{Rs } 1920$.

Anil bought 12 toys, so CP per toy = $1920 / 12 = \text{Rs } 160$.

2. Calculate the Labeled Price (LP) per toy.

Let the LP be M. He sells 8 toys at a 20% discount: $SP1 = 8 \times M \times (1 - 0.20) = 6.4M$.

He sells 4 toys at a further 25% discount on the discounted price (0.8M): $SP2 = 4 \times (0.8M) \times (1 - 0.25) = 2.4M$.

Total SP = $SP1 + SP2 = 6.4M + 2.4M = 8.8M$.

$8.8M = 2112 \rightarrow M = 2112 / 8.8 = 240$. The labeled price per toy is Rs 240.

3. Calculate the profit percentage with no discounts.

If sold with no discount, the SP per toy would be the LP, i.e., Rs 240.

CP per toy = Rs 160. Profit per toy = $SP - CP = 240 - 160 = \text{Rs } 80$.

Profit Percentage = $(\text{Profit} / \text{CP}) \times 100 = (80 / 160) \times 100 = 50\%$.

Correct Answer: A

Question 16

If a seller gives a discount of 15% on retail price, she still makes a profit of 2%. Which of the following ensures that she makes a profit of 20%?

A) Give a discount of 5% on retail price.

B) Give a discount of 2% on retail price.

C) Increase the retail price by 2%.

✓ D) Sell at retail price. ← **Correct Answer**

Solution:

Step 1: Relate Retail Price (RP) and Cost Price (CP).

Let the Cost Price be CP and the Retail Price be RP.

In the first scenario, a 15% discount is given on RP: $SP1 = RP \times (1 - 0.15) = 0.85 \times RP$.

This SP leads to a 2% profit: $SP1 = CP \times (1 + 0.02) = 1.02 \times CP$.

Equating: $0.85 \times RP = 1.02 \times CP \rightarrow RP = (102 / 85) \times CP = (6/5) \times CP = 1.2 \times CP$.

This tells us the retail price is marked up by 20% over the cost price.

Step 2: Determine the selling price for a 20% profit.

To make a 20% profit: $SP2 = CP \times (1 + 0.20) = 1.2 \times CP$.

Step 3: Compare the required selling price with the retail price.

From Step 1, $RP = 1.2 \times CP$. From Step 2, $SP2 = 1.2 \times CP$. Therefore, $SP2 = RP$.

To achieve a 20% profit, the item must be sold at its retail price (zero discount).

Step 4: Evaluate the options.

(D) Sell at retail price: $SP = RP = 1.2 \times CP$ (20% profit). This matches our requirement.

Correct Answer: D

Question 17

In a market, the price of medium quality mangoes is half that of good mangoes. A shopkeeper buys 80 kg good mangoes and 40 kg medium quality mangoes from the market and then sells all

these at a common price which is 10% less than the price at which he bought the good ones. His overall profit is

- A) 6%
- ✓ B) 8% ← Correct Answer
- C) 10%
- D) 12%

Solution:

Step 1: Define prices.

Let the price of good quality mangoes be Rs $2P$ per kg. Then, the price of medium quality mangoes is Rs P per kg.

Step 2: Calculate the Total Cost Price (CP).

Shopkeeper buys 80 kg of good mangoes and 40 kg of medium mangoes.

Cost of good mangoes = $80 \text{ kg} \times 2P/\text{kg} = 160P$. Cost of medium mangoes = $40 \text{ kg} \times P/\text{kg} = 40P$.

Total CP = $160P + 40P = 200P$.

Step 3: Calculate the Total Selling Price (SP).

Total quantity of mangoes sold = $80 + 40 = 120 \text{ kg}$.

The common selling price is 10% less than the price of good mangoes: SP per kg = $2P \times (1 - 0.10) = 1.8P$.

Total SP = Total quantity $\times$ SP per kg = $120 \times 1.8P = 216P$.

Step 4: Calculate the overall profit percentage.

Profit = Total SP - Total CP = $216P - 200P = 16P$.

Profit Percentage = $(\text{Profit} / \text{Total CP}) \times 100 = (16P / 200P) \times 100 = 8\%$.

Correct Answer: B

Question 18

If Fatima sells 60 identical toys at a 40% discount on the printed price, then she makes 20% profit. Ten of these toys are destroyed in fire. While selling the rest, how much discount should be given on the printed price so that she can make the same amount of profit?

- A) 30%
- B) 25%
- C) 24%
- ✓ D) 28% ← Correct Answer

Solution:

Step 1: Determine the Cost Price (CP) per toy.

Let the Printed Price (PP) be Rs 100 for simplicity.

A 40% discount means the Selling Price (SP) is $100 \times (1 - 0.40) = \text{Rs } 60$.

At this SP, Fatima makes a 20% profit. $SP = CP \times (1 + 0.20) \rightarrow 60 = CP \times 1.2 \rightarrow CP = 60 / 1.2 = \text{Rs } 50$.

Step 2: Calculate the total profit to be made.

Profit per toy = $SP - CP = 60 - 50 = \text{Rs } 10$. Total target profit = $60 \text{ toys} \times \text{Rs } 10/\text{toy} = \text{Rs } 600$.

Step 3: Calculate the required revenue from the remaining toys.

Total cost of the initial 60 toys = $60 \times CP = 60 \times 50 = \text{Rs } 3000$.

10 toys are destroyed, leaving 50 toys to be sold.

Total Revenue = Total Cost + Total Profit = $3000 + 600 = \text{Rs } 3600$.

Step 4: Calculate the new selling price and the required discount.

New SP per toy = Total Revenue / 50 = $3600 / 50 = \text{Rs } 72$.

The Printed Price is still Rs 100. Discount = $PP - \text{New SP} = 100 - 72 = \text{Rs } 28$.

Discount Percentage = $(\text{Discount} / PP) \times 100 = (28 / 100) \times 100 = 28\%$.

Correct Answer: D

Question 19

A wholesaler bought walnuts and peanuts, the price of walnut per kg being thrice that of peanut per kg. He then sold 8 kg of peanuts at a profit of 10% and 16 kg of walnuts at a profit of 20% to a shopkeeper. However, the shopkeeper lost 5 kg of walnuts and 3 kg of peanuts in transit. He then mixed the remaining nuts and sold the mixture at Rs. 166 per kg, thus making an overall profit of 25%. At what price, in Rs. per kg, did the wholesaler buy the walnuts?

✓ A) 96 ← Correct Answer

B) 98

C) 86

D) 84

Solution:

Let the wholesaler's cost price for peanuts be P Rs/kg. Then, the wholesaler's cost price for walnuts is $W = 3P$ Rs/kg.

Step 1: Calculate the shopkeeper's total cost price (CP_S)

Price of peanuts for shopkeeper = $P \times (1 + 10/100) = 1.1P$.

Price of walnuts for shopkeeper = $3P \times (1 + 20/100) = 3P \times 1.2 = 3.6P$.

The shopkeeper buys 8 kg of peanuts and 16 kg of walnuts.

Cost of peanuts for shopkeeper = $8 \times 1.1P = 8.8P$. Cost of walnuts for shopkeeper = $16 \times 3.6P = 57.6P$.

Total CP for the shopkeeper = $8.8P + 57.6P = 66.4P$.

Step 2: Calculate the shopkeeper's total selling price (SP_S)

Remaining walnuts = $16 - 5 = 11$ kg. Remaining peanuts = $8 - 3 = 5$ kg. Total remaining mixture = $11 + 5 = 16$ kg.

Total SP for the shopkeeper = $16 \text{ kg} \times 166 \text{ Rs/kg} = \text{Rs. } 2656$.

Step 3: Use the shopkeeper's profit to find P

The shopkeeper makes an overall profit of 25%: $SP_S = CP_S \times (1 + 25/100)$.

$2656 = 66.4P \times 1.25 = 83P \rightarrow P = 2656 / 83 = 32$.

The wholesaler's cost price for peanuts is Rs. 32 per kg.

Step 5: Calculate the price of walnuts for the wholesaler

$$W = 3P = 3 \times 32 = \text{Rs. } 96 \text{ per kg.}$$

Correct Answer: A

Question 20

On selling a pen at 5% loss and a book at 15% gain, Karim gains Rs. 7. If he sells the pen at 5% gain and the book at 10% gain, he gains Rs. 13. What is the cost price of the book in Rupees?

A) 95

B) 85

✓ C) 80 ← Correct Answer

D) 100

Solution:

Let the cost price of the pen be P and the book be B.

1. Formulate equations from the given conditions:

Condition 1: 5% loss on pen, 15% gain on book, net gain Rs. 7.

$$-0.05P + 0.15B = 7 \dots(1)$$

Condition 2: 5% gain on pen, 10% gain on book, net gain Rs. 13.

$$+0.05P + 0.10B = 13 \dots(2)$$

2. Solve the system of linear equations:

Add equation (1) and equation (2): $(-0.05P + 0.05P) + (0.15B + 0.10B) = 7 + 13$.

$$0.25B = 20 \rightarrow B = 20 / 0.25 = 80.$$

The cost price of the book is Rs. 80.

Correct Answer: C

Question 21

A shopkeeper sells two tables, each procured at cost price p, to Amal and Asim at a profit of 20% and at a loss of 20%, respectively. Amal sells his table to Bimal at a profit of 30%, while Asim sells his table to Barun at a loss of 30%. If the amounts paid by Bimal and Barun are x and y, respectively, then $(x - y) / p$ equals

✓ A) 1 ← Correct Answer

B) 1.2

C) 0.50

D) 0.7

Solution:

1. Determine the cost price for Amal and Asim.

Shopkeeper's cost price = p.

Amal buys at 20% profit, so Amal's CP = $p(1 + 0.20) = 1.2p$.

Asim buys at 20% loss, so Asim's CP = $p(1 - 0.20) = 0.8p$.

2. Determine the price paid by Bimal and Barun.

Bimal buys from Amal. Amal sells at a 30% profit on his CP.

$x = \text{Amount paid by Bimal} = (\text{Amal's CP}) \times (1 + 0.30) = (1.2p) \times 1.3 = 1.56p$.

Barun buys from Asim. Asim sells at a 30% loss on his CP.

$y = \text{Amount paid by Barun} = (\text{Asim's CP}) \times (1 - 0.30) = (0.8p) \times 0.7 = 0.56p$.

3. Calculate the required expression.

$(x - y) / p = (1.56p - 0.56p) / p = 1.00p / p = 1$.

Correct Answer: A

Question 22

Mukesh purchased 10 bicycles in 2017, all at the same price. He sold six of these at a profit of 25% and the remaining four at a loss of 25%. If he made a total profit of Rs. 2000, then his purchase price of a bicycle, in Rupees, was

A) 6000

B) 8000

✓ C) 4000 ← **Correct Answer**

D) 2000

Solution:

Let 'c' be the purchase price of one bicycle. Total CP = $10c$.

Calculate total revenue (Total SP):

6 bicycles sold at 25% profit: $SP = 6 \times (c \times 1.25) = 7.5c$.

4 bicycles sold at 25% loss: $SP = 4 \times (c \times 0.75) = 3c$.

Total SP = $7.5c + 3c = 10.5c$.

Use the profit information to find c:

Total Profit = Total SP - Total CP $\rightarrow 2000 = 10.5c - 10c \rightarrow 2000 = 0.5c \rightarrow c = 4000$.

The purchase price of a bicycle is Rs. 4000.

Correct Answer: C

Question 23

Minu purchases a pair of sunglasses at Rs.1000 and sells to Kanu at 20% profit. Then, Kanu sells it back to Minu at 20% loss. Finally, Minu sells the same pair of sunglasses to Tanu. If the total profit made by Minu from all her transactions is Rs.500, then the percentage of profit made by Minu when she sold the pair of sunglasses to Tanu is

A) 35.42%

B) 52%

✓ C) 31.25% ← Correct Answer

D) 26%

Solution:

Transaction 1: Minu to Kanu

Minu's Cost Price (CP1) = 1000. Minu sells at 20% profit: Selling Price (SP1) = $1000 \times 1.20 = 1200$.

Minu's profit (Profit1) = $1200 - 1000 = 200$.

Transaction 2: Kanu to Minu

Kanu's cost price was 1200. Kanu sells at 20% loss: Selling Price = $1200 \times 0.80 = 960$.

Minu buys it back at this price. So, Minu's new cost price (CP2) = 960.

Transaction 3: Minu to Tanu

Minu's total profit from all her transactions is 500.

Total Profit = Profit1 + Profit2 = $200 + \text{Profit2} = 500 \rightarrow \text{Profit2 (from sale to Tanu)} = 300$.

Profit Percentage for this sale = $(\text{Profit2} / \text{CP2}) \times 100 = (300 / 960) \times 100 = (5 / 16) \times 100 = 31.25\%$.

Correct Answer: C

Question 24

Gopi marks a price on a product in order to make 20% profit. Ravi gets 10% discount on this marked price, and thus saves Rs 15. Then, the profit, in rupees, made by Gopi by selling the product to Ravi, is

✓ A) 10 ← Correct Answer

B) 25

C) 15

D) 20

Solution:

Define variables:

Let CP be the Cost Price. Let MP be the Marked Price. Let SP be the Selling Price.

Use the given information to form equations:

Gopi marks the price to make 20% profit: $\text{MP} = \text{CP} + 0.20 \times \text{CP} = 1.2 \times \text{CP}$.

Ravi gets a 10% discount on the MP. The discount amount is $0.10 \times \text{MP}$.

This discount saves Ravi Rs 15: Discount Amount = 15.

Solve for MP, CP, and SP:

From the discount information: $0.10 \times \text{MP} = 15 \rightarrow \text{MP} = 150$.

Now find the CP using the markup relation: $150 = 1.2 \times \text{CP} \rightarrow \text{CP} = 150 / 1.2 = 125$.

The Selling Price (SP) is the price after the discount: $\text{SP} = \text{MP} - \text{Discount} = 150 - 15 = 135$.

Calculate Gopi's profit:

Profit = $\text{SP} - \text{CP} = 135 - 125 = 10$. The profit made by Gopi is Rs 10.

Correct Answer: A